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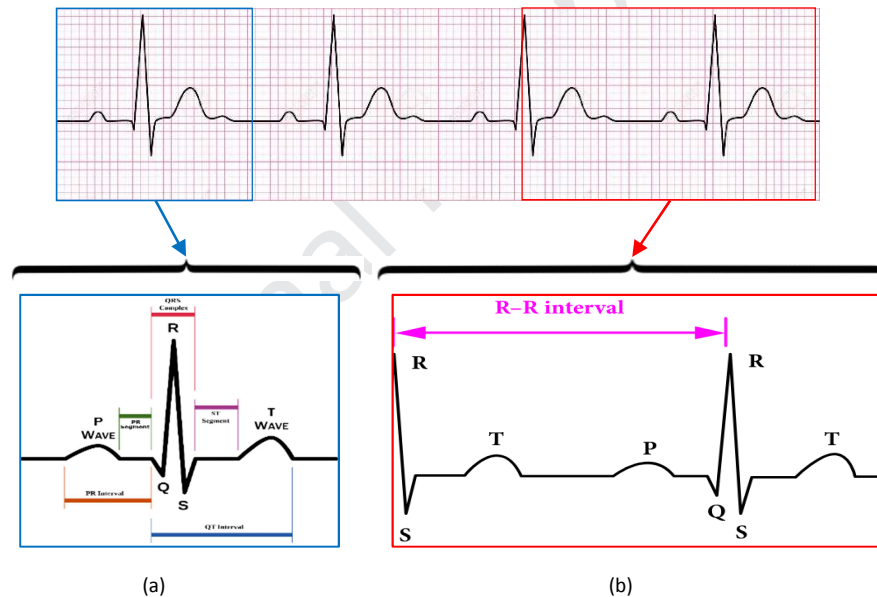


Fig. 1. Heartbeat segmentation from a typical electrocardiogram signal. (a) Typical heartbeat segmentation, (b) proposed heartbeat segmentation.



Electrocardiogram Heartbeat Classification Based on a Deep Convolutional Neural Network and Focal Loss

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Abstract

The electrocardiogram (ECG) is an effective tool for cardiovascular disease diagnosis and arrhythmia detection. Most methods proposed in the literature include the following steps: 1) denoizing, 2) segmentation into heartbeats, 3) feature extraction, and 4) classification. In this paper, we present a deep learning method based on a convolutional neural network (CNN) model. CNN models can perform feature extraction automatically and jointly with the classification step. In other words, our proposed method does not require a feature extraction step with hand-crafted techniques. Our proposed method is also based on an algorithm for heartbeat segmentation that is different from most existing methods. In particular, the segmentation algorithm defines each ECG heartbeat to start at an R-peak and end after 1.2 times the median RR time interval in a 10-s window. This method is simple and effective, as it does not use any form of filtering or processing that requires assumptions about the signal morphology or spectrum. Although enhanced ECG heartbeat classification algorithms have been proposed in the literature, they failed to achieve high performance in detecting some heartbeat categories, especially for imbalanced datasets. To overcome this challenge, we propose an optimization step for the deep CNN model using a novel loss function called focal loss. This function focuses on minority heartbeat classes by increasing their importance. We trained and evaluated our proposed model with the MIT-BIH and INCART datasets to identify five arrhythmia categories (N, S, V, Q, and F) based on the Association for Advancement of Medical Instrumentation (AAMI) standard. The evaluation results revealed that the focal loss function improved the classification accuracy for the minority classes as well as the overall metrics. Our proposed method achieved 98.41% overall accuracy, 98.38% overall F1-score, 98.37% overall precision, and 98.41% overall recall. In addition, our method achieved better performance than that of existing state-of-the-art methods.

Keywords: ECG Classification; AAMI standard; Convolutional Neural Network; Focal loss; Heartbeat category; Imbalanced data; INCART; MIT-BIH.

1. Introduction

An electrocardiogram (ECG) signal is an electrical signal emitted by the heart that can be measured non-intrusively by sensors attached to a patient's chest [1]. This signal contains a large amount of pathological information about the patient's heart processes. A heart arrhythmia [2] is among the main types of cardiovascular disease. An arrhythmia is defined as an abnormal heartbeat (too fast, too slow, too early, or irregular). Various heart arrhythmias are harmless, such as tachycardia and bradycardia. If arrhythmias are significantly abnormal or caused by a damaged heart, they can be dangerous and even fatal. Arrhythmias can be detected by the inspection and classification of ECG heartbeats [3]. A plethora of studies have focused on the development of heartbeat classification algorithms [4,5].

Most recent works in the literature present solutions based on deep convolutional neural network (CNN) models. CNNs are used in various fields, including medicine, owing to their high performance in image classification and object detection [6]. Fundamentally, a CNN is similar to a multilayer perceptron (MLP) network [7], in which each neuron is connected to several neurons in the previous layer. The neuron computes a weighted sum of all its incoming neurons and then applies a non-linear activation function that makes neural networks capable of modeling complex learning problems. The MLP network can be extended to a deep MLP if many hidden layers are used. In the same way, a CNN is considered an MLP network with a specific structure. The CNN network has an input layer, an output layer, and various hidden layers, such as a convolutional layer, pooling layer, and fully connected layer. Each hidden layer performs different tasks on the output from the preceding layer. The convolutional layers are the main component of the CNN, containing a set of learnable filters



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applied to the input image or to other feature maps in the network. Each filter is convolved through the height and width of the input data to produce a new feature or activation map of the data. The pooling layers resemble convolutional layers. However, they perform special tasks, such as max pooling or average pooling (i.e., considering the maximum or average value in a certain filter region). These layers are usually applied to decrease the CNN dimensions. The fully connected layers are similar to the output layer of the MLP network and are used before the output layer to flatten the feature maps into one vector.

One problem with ECG datasets is the high imbalance between different classes. In general, there is some degree of imbalance in any dataset [8]. The performance of the model is not significantly affected when the degree of imbalance is relatively low. However, in medical diagnosis, and particularly for ECG signals, data are highly imbalanced because samples exhibiting disease are more limited than normal samples. Dataset imbalance can cause two problems. First, the training becomes ineffective, as most observations are easy samples (normal samples) that provide no learning benefit to the model. Second, normal samples can dominate the training and cause the classifier to favor classes that have a large number of labeled samples. A common solution to the imbalanced class problem involves augmenting small classes with more data through replication or transformation operations, such as reflection and rotation. However, we believe that augmenting the amount of data is not always suitable because it significantly increases the computational costs and introduces new training data (through transformations) that may be very different from the data to be really tested.

A major objective of a classification model is to minimize a defined loss function. However, as the famous expression says, "you get what you measure", if the loss function considers only the classification accuracy, the model may focus on majority classes because it will attempt to minimize the metrics. For example, in a binary classification scenario, if the majority class comprises 99% of the learning data, then simply predicting that all samples belong to this class will make the accuracy reach 99%. In fact, this is the best possible classifier; however, a problem remains with minority classes. Thus, to make the model applicable to minority classes, we must use a loss function in which the cost of misclassifying the minority class is much higher than that of misclassifying the majority class. Recently, researchers have proposed a loss function called focal loss that has this property [9]. To our knowledge, no previous studies have applied focal loss to the problem of ECG heartbeat classification.

In this work, we present a deep learning solution to classify ECG heartbeats into five classes (N, S, V, F, and Q) as defined by the Association for Advancement of Medical Instrumentation (AAMI) standard [10]. Our first contribution consists of designing a custom CNN model with two convolutional blocks. Each block is composed of three one-dimensional (1D) convolutional layers that are activated by a leaky rectified linear unit (ReLU) function [11]. Batch normalization and dropout layers follow each block. The solution to the imbalanced data problem constitutes the second main contribution of this work through the use of a focal loss function [9]. In addition, an atypical segmentation technique on the ECG records is used to enrich the features embedded in each segmented heartbeat. Improved results are expected from this work due to the following contributions:

1. Employing a new segmentation method to separate ECG signals into heartbeats that does not require any form of filtering or processing that makes assumption about the signal morphology or spectrum.
2. Developing a deep CNN model using the latest state-of-the-art architectures and training techniques for the classification of ECG recordings.
3. Exploiting a novel type of loss function to effectively handle the imbalanced class problem.

The remainder of this paper is structured as follows. Section 2 surveys the state-of-the-art of ECG signal classification for heart disease diagnosis. Section 3 describes the architecture of our CNN model, including algorithms and methods used to reduce the imbalanced data problem and enhance the performance of the model. Section 4 describes the simulation environment and presents the performance metrics, experimental results, and a discussion. It also presents a performance comparison of the proposed solution with state-of-the-art works. Finally, Section 5 concludes the paper.

2. Related Work

ECG heartbeat classification has been heavily studied in the literature, and high-quality reviews are provided in [12-15]. In this section, we review the most important methods developed for ECG signal classification. For instance, one of the earliest methods was developed by Castillo et al. [4], who proposed a method based on the fusion of three classifiers using a Mamdani-type fuzzy inference system. The three fused classifiers were the fuzzy k-nearest neighbors, MLP with gradient descent and momentum back-propagation, and MLP with scaled conjugate gradient back-propagation. This method achieved a high classification accuracy rate of 98%. Melin et al. [16,17] proposed a new model for arrhythmia classification using a modular learning vector quantization (LVQ) network architecture, and proposed a type-2 fuzzy system. The authors focused



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on the implementation of the type-2 fuzzy system to determine the shortest distance in an LVQ competitive layer. In their study, the MIT-BIH arrhythmia database with 15 classes was used. The results demonstrated that using five modules was an effective approach for arrhythmia classification. The study conducted by Martis et al. [18] was based on the discrete wavelet transform (DWT) and the support vector machine (SVM) classifier.

Li et al. [19] used the Daubechies wavelet packet entropy (WPE) and random forest for feature extraction and ECG signal classification. They applied wavelet packet decomposition with six levels to decompose ECG signals into representative features to calculate the entropy. Then, they used the random forest classifier to identify heartbeat categories.

Garcia et al. [20] presented a new ECG representation founded on the vectorcardiogram called the temporal vectorcardiogram, which is an electrical activity in a three-dimensional plot. To extract features from this plot, the authors used complex networks in which the parameters were estimated by applying the particle swarm optimization (PSO) algorithm. This algorithm makes it possible to achieve optimal complex network parameters that are joint with the RR interval and entered as input to the SVM classifier to classify the heartbeat and detect arrhythmia categories.

Acharya et al. [21] proposed a classification algorithm with a nine-layer deep CNN and sequence-to-sequence modeling to automatically detect five different categories of heartbeats (N, S, V, F, Q). They used synthetic data to overcome the dataset imbalance problem. The examples of the augmented data were generated after pre-processing by varying the standard deviation and mean of the Z-score, which was calculated from the original normalized ECG records.

Hammada et al. [22] developed a fast classifier based on rules related to 13 ECG-extracted features to classify heartbeats into normal and abnormal classes. First, they denoised ECG signals using a band-pass Butterworth filter. Then, they applied the Pan-Tompkins algorithm to extract the ECG characteristics required to classify the heartbeats. The authors used ECG signals from the first channel (MLII) of the MIT-BIH arrhythmia dataset, which were classified with SVM, MLP, and a feed-forward neural network and compared with the proposed classifier. They applied their method on the PTB dataset and obtained results revealing an overall sensitivity of 99.98% and a positive predictive value of 99.99%.

Amrani et al. [23] proposed a deep CNN of up to 19 weight layers with Q-Gaussian multi-class SVM to detect four heartbeat categories (normal heartbeats, atrial fibrillation, atrial flutter, and paroxysmal supraventricular tachycardia). The proposed classifier was applied on ECG signals from three cardiac databases: the MIT-BIH arrhythmia database (MIT-BIH), atrial fibrillation database, and MIT-BIH supraventricular arrhythmia database, where 80% of the signals were used for training and 20% were used for testing. The authors also used the multi-canonical correlation analysis technique without noise filtering or pre-processing methods to improve the training task speed and increase the accuracy from 94.74% to 97.37%.

Ramirez et al. [24] developed a method that combined two neural network models for two different ECG leads. The two models were combined at the decision level using two types of fuzzy logic techniques. The final classification accuracy was improved to 93.80% from 92.90% and 92.70% for the two basic modules using lead 1 and lead 2, respectively.

Yildirim et al. [25] developed a convolutional auto-encoder (CAE) model and a long short-term memory (LSTM) classifier to detect five arrhythmia classes: atrial premature beat, left bundle branch block, normal sinus rhythm, right bundle branch block, and premature ventricular contraction. The proposed 16-layer deep CAE was based on a non-linear compression structure to compress and reduce the large ECG signal from 260 samples into 32 samples per segment with minimum loss. LSTM, which is a special type of recurrent neural network (RNN) with five-unit dense layers, was then created to classify the compressed signals using the coded ECG features. The authors used 70%, 15%, and 15% of MIT-BIH ECG signals to train, validate, and test their model, respectively. They achieved an average accuracy, precision, recall, and F1-score of 99%. The limitations of the proposed system are the complexity of the deep learning compression model and the long time required to obtain the coded ECG features.

Marinho et al. [26] deployed the fast Fourier transform, Goertzel, higher-order statistics (HOS), and structural co-occurrence matrix as feature extraction techniques. Then, they applied naive Bayes (NB), SVM, MLP, and optimum path forest (OPF) to classify heartbeats into five categories corresponding to the AAMI standard (N, S, V, F, and Q). Using all combinations of the listed feature extraction techniques and classifiers, the best results in the identification of cardiac arrhythmia categories were obtained with the HOS extractor method and NB classifier. This combination achieved an accuracy of 94.3%.

Sannino et al. [27] proposed a method based on a deep neural network (DNN) method to classify ECG heartbeats. However, their method began with a heavy pre-processing step consisting of denoizing, segmentation, resampling, and temporal feature extraction. The DNN performed classification based on seven fully connected hidden layers with 5, 10, 30, 50, 30, 10, and 5 neurons each. The authors conducted experiments on the well-known MIT-BIH arrhythmia database. They claimed to achieve 99% accuracy but did not provide details on how they trained their DNN or what parameters they used.

Mousavi et al. [28] developed an automatic heartbeat classifier using the sequence-to-sequence model and a deep CNN. The sequence-to-sequence model was composed of an RNN encoder and decoder with LSTM to compute the class of



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each heartbeat of the input sequence. The authors applied their solution on the MIT-BIH database and achieved an accuracy of 96.18%.

Kachuee et al. [29] introduced a method for transferring knowledge from one CNN trained on an ECG dataset to another ECG dataset. The method can classify five different arrhythmias in accordance with the AAMI standard. The authors trained the proposed CNN model on five classes of the MIT-BIH database and then transferred the knowledge learned by the CNN to solve the two-class problem of PTB diagnostic datasets.

The authors of [30] developed and implemented an effective system for ECG classification based on the faster region-based CNN (FR-CNN) algorithm. The study used the MIT-BIH database in addition to 1D ECG signals consisting of preprocessed patient signals. The FR-CNN was used to transform 1D ECG signals into a two-dimensional image for experimental training and testing using five categories.

More recently, Li et al. [31] used a deep residual network (ResNet) for the classification of cardiac arrhythmias. Their CNN was composed of four residual blocks, each consisting of three 1D convolutional layers, three batch normalization layers, three ReLU layers, and an identity shortcut connection structure. The authors applied their CNN to the MIT-BIH dataset and achieved an average accuracy, sensitivity, and positive predictivity of 99.06%, 93.21%, and 96.76%, respectively.

A study by Wang et al. [32] proposed a dual fully connected neural network model for the accurate classification of heartbeats as an automatic arrhythmia detection system. This model followed the AAMI inter-patient standard including five classes (N, S, V, F, and Q). It began by extracting 105 features from preprocessed ECG signals. Then, it introduced a two-layer fully connected neural network classifier. This signifies that the method did not exploit the full power of deep learning, as it used a separate hand-crafted feature extraction step. For verification, both the MIT-BIH and supraventricular arrhythmia datasets were used. The proposed model demonstrated high performance for arrhythmia detection, with a particularly high sensitivity for S and V classes.

Pandey et al. [33] developed an expert system based on LSTM to automatically detect abnormalities in heart rhythms. In their study, 45 cardiac features were extracted from ECG signals based on HOS, wavelets, morphological descriptors, and RR intervals. These extracted features were used as a sequence to a single LSTM model to classify five distinct arrhythmic rhythms. Again, this signifies that the method did not exploit the full power of deep learning. The performance evaluation of the model achieved 99.37% accuracy, 96.73% precision, 99.14% specificity, 94.89% sensitivity, and 95.77% F1-score.

In [34], Chen et al. presented a method called MF-CBRNN that contained two parallel hybrid branches to concentrate on beat-based information in the heartbeats and segment-based information in the adjacent segments of these beats. Each heartbeat provided morphological information, and the adjacent intervals of the beat augmented the temporal information. In addition, in each branch, the authors of the paper used the CNN and bidirectional LSTM (BLSTM) to extract temporal and morphological information. Then, the captured features from each branch were fused before passing through the classification layers. The proposed model yielded an accuracy of 99.56% and an F1-score of 96.40% under the intra-patient paradigm.

Unlike most recent methods that use deep neural networks, Bidias et al. [35] used the conditional entropy of ordinal patterns (CEOP) and the permutation entropy (PE) to classify heartbeats. They regulated their threshold formula parameters with the European Society of Cardiology ST-T and MIT-BIH arrhythmia datasets. Their experiments demonstrated that the system was not very sensitive to these parameter settings, and the results of both methods (PE and CEOP) were similar. The authors achieved a classification rate of 93.62% for the MIT-BIH database.

Similar to the study conducted by Bidias et al. [35], Zhang et al. [36] recently proposed a method based on the SVM classifier. The proposed method selected the most important features, such as intra-beat and inter-beat intervals, amplitude morphology, and area morphology, to enable the high-level prediction of classes. To this end, the authors used a one-versus-one (OvO) feature ranking method and a feature search technique with an OvO-rule SVM binary classifier.

3. Method

3.1. ECG segmentation into heartbeats

The study by Kachuee et al. [29] was unique in that it proposed an interesting new approach for segmenting ECG signals into heartbeats, which was also used by Walsh et al. in [37]. Because one ECG record is a signal recorded over several minutes or even hours, it is composed of hundreds of heartbeats. Therefore, before implementing a heartbeat



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classification algorithm, the ECG signal must be segmented into single heartbeats. More than 30 segmentation algorithms have been proposed in the literature [38]. The most famous algorithm is known as the Pan–Tompkins algorithm and its variants [39,40]. The Pan–Tompkins algorithm relies on detecting the most striking peak and its neighboring waveform structure in an ECG signal called the QRS complex. Figure 1 presents a plot of a typical ECG record and illustrates the structure of one ECG heartbeat including the QRS complex waveforms. Table 1 summarizes the most common value ranges of the ECG parameters.

Table 1. Common value ranges of electrocardiogram (ECG) signal parameters.

ECG Parameter	Duration (s)	Amplitude (mv)
P wave	0.08–0.1	0.25
T-wave	0.16–0.2	Tamp > 0
RR Interval	0.6–1.2	-
PR Interval	0.12–0.22	Ramp > 0
ST Interval	0.2–0.32	Isoelectric: 0
QT Interval	0.35–0.45	-
QRS Complex	0.08–0.1	Qam < 0, Ram > 0, Sam < 0, Upper amplitude (2.5–3)

Once the QRS complex is detected, a heartbeat is defined as the waveform that starts before the P wave and ends just after the T-wave. Most existing ECG classification methods segment heartbeats as illustrated in Fig. 1, where the QRS complex and R-peak are at the middle of the heartbeat.

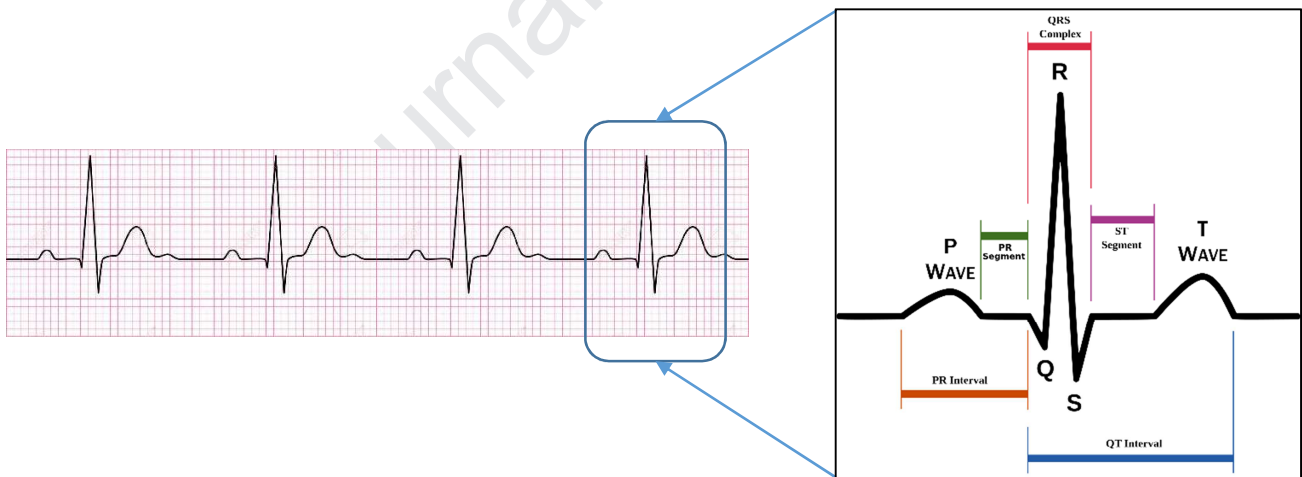


Fig. 1. Typical electrocardiogram (ECG) signal (left) and typical ECG heartbeat with QRS complex (right).

Figure 2(b) illustrates the proposed segmentation, where each heartbeat is defined to start at an R-peak and end after 1.2 times the median RR time interval in a 10-s window. This is in contrast to the existing method of segmentation illustrated in Fig. 2(a), where a heartbeat is defined to start at some PR time interval before an R-peak and end at some time after the T-wave. The proposed heartbeat extraction technique is effective because it includes important features, such as RR intervals, the PR interval, and even the QT interval. These features are commonly used in many classification methods in the literature [12–15] because they are effective in class differentiation. In addition, the proposed segmentation method does not require resampling techniques to make the heartbeats the same length. The extracted heartbeats have equal lengths and are suitable as input to the CNN model.

This segmentation method is used in the current study. The annotation file for each ECG signal contains the sample location of the annotation according to the AAMI standard. Under the new segmentation, each heartbeat starts at an R-peak



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and should cover the next R-peak and beyond. The annotation assigned to the segmented heartbeat is given by the second R-peak.

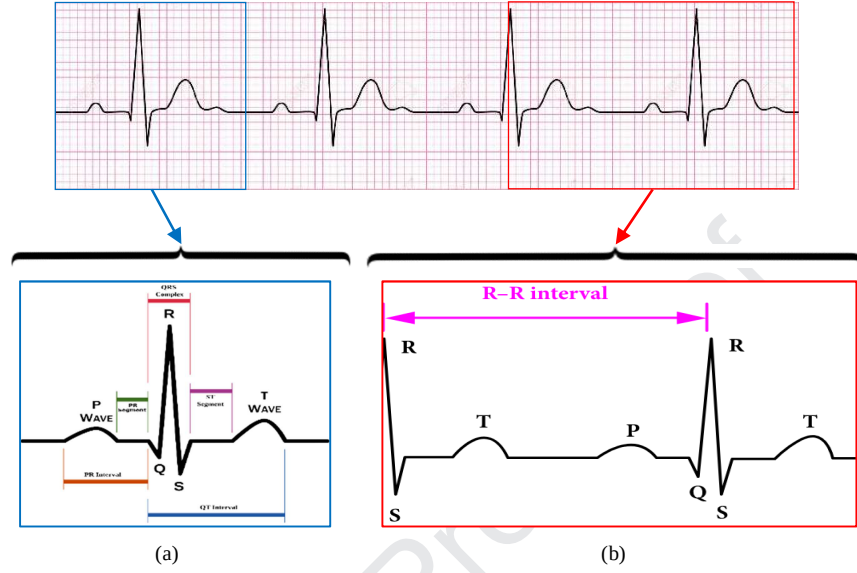


Fig. 2. Heartbeat segmentation from a typical electrocardiogram signal. (a) Typical heartbeat segmentation, (b) proposed heartbeat segmentation.

3.2. Focal loss to address imbalanced data

Focal loss was proposed by Tsung-Yi et al. [41] as a solution for imbalanced data [42] for dense object detection. It allows for a high training accuracy with an imbalance ratio of 1:1000 scale between foreground and background classes. The authors used the focal loss technique to improve one-stage object detectors and train the RetinaNet detector. The focal loss considers the contribution of each sample to the loss based on the classification error. Using this function, the loss decreases when a sample is correctly classified by the model. This approach solves the class imbalance problem by making the loss indirectly focus on challenging classes. To introduce focal loss, we introduce the cross-entropy (CE) loss function [43] for binary classification, which is expressed as follows:

$$CE(p, y) = \begin{cases} -\log(p) & \text{if } y = 1 \\ -\log(1 - p) & \text{otherwise} \end{cases} \quad (1)$$

In Eq. (1), $p \in [0, 1]$ is the model's estimated probability and $y \in \{\pm 1\}$ is the ground truth class. Focal loss can be extended to the multi-class case by defining parameter p_t as follows:

$$p_t = \begin{cases} p & \text{if } y = 1 \\ 1 - p & \text{otherwise} \end{cases} \quad (2)$$

Corresponding to Eq. (2), the CE function is expressed as follows:

$$CE(p, y) = CE(p_t) = -\log(p_t) \quad (3)$$

We now introduce the balanced CE loss function, as illustrated in Eq. (4). This addresses the class imbalance by adding a weighting factor α for class 1 and $1 - \alpha$ for class -1.

$$CE(p_t) = -\alpha_t \log(p_t) \quad (4)$$



This formula is considered a simple extension to CE, where α can be set by the inverse of the class frequency or adapted as a hyper parameter fixed by cross-validation. Focal loss, denoted L_f , is an extension of the CE loss function by including a weighted term, as given in Eq. (5), where α and γ are two adjustable parameters:

$$L_f = -\alpha_t (1 - P_t)^\gamma \log(P_t), \quad (5)$$

where

$$\alpha_t = \begin{cases} \alpha & \text{if } y = 1 \\ 1 - \alpha & \text{otherwise,} \end{cases} \quad (6)$$

where the first parameter γ is a fixed positive value that regulates the rate at which samples are down-weighted. If $\gamma = 0$, the focal loss function resembles a CE function. When γ increases, the efficiency of the modulating factor increases as well (several experiments have provided $\gamma = 2$ as the optimal value). Moreover, α is used as a fixed value between 0 and 1 to balance positive and negative labeled samples. This parameter constitutes a common solution for balancing classes. The classification accuracy using the α -balanced form is superior to that using the non- α -balanced form.

3.3. Architecture of proposed CNN model

Our classification model, as presented in Fig. 3, is based on a CNN for classification. The proposed CNN model is composed of two convolutional blocks. Each block is composed of three convolutional (1D) layers activated by the leaky ReLU activation function [11]. The convolution operation is calculated using Eq. (7) as follows:

$$y_n = \sum_{k=0}^{N-1} x_k f_{(n-k)}, \quad (7)$$

where N , x , and f represent the vector dimension, input signal, and filter, respectively. Vector y represents the output signal. Each of the 1D convolutional layers has 256 filters used to extract additional features from the input signal. The size of each filter is 2, and the stride is equal to 1. After every convolutional block, batch normalization [44] and dropout [45] operations are applied in the model. Batch normalization is used to normalize the data by scaling and adjusting the activations. This function improves the stability and performance of the classifier and accelerates its learning with the gradient descent algorithm. Batch normalization is also applied to resolve the internal covariate shift in which the activation distribution always varies during the training process [44]. This variation greatly disturbs the training because each convolutional layer must be adapted to the new distribution for every training epoch. During the training process, the batch normalization method performs a list of operations. First, it calculates the batch mean μ_B and standard deviation σ_B of the mini-batch of the input. These parameters are expressed by Eqs. (8) and (9), respectively, where the input is a list of x values over batch $B = \{x_1 \dots x_m\}$.

$$\mu_B = \frac{1}{m} \sum_{i=1}^m x_i \quad (8)$$

$$\sigma_B^2 = \frac{1}{m} \sum_{i=1}^m (x_i - \mu_B)^2 \quad (9)$$

Second, batch normalization normalizes the input layer with the calculated batch mean and variance using Eq. (10):

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad (10)$$

where ϵ is a very small value added to prevent division by zero. Finally, batch normalization shifts and scales the input data to obtain the output y_i using Eq. (11):



$$y_i = \gamma \hat{x}_i + \beta \quad (11)$$

Here, γ and β are the learned parameters. After the batch normalization operations, a dropout method is used to prevent the network from overfitting by ignoring a randomly selected neuron through training. Dropout is beneficial in large networks with complex fitting and few labeled samples [45].

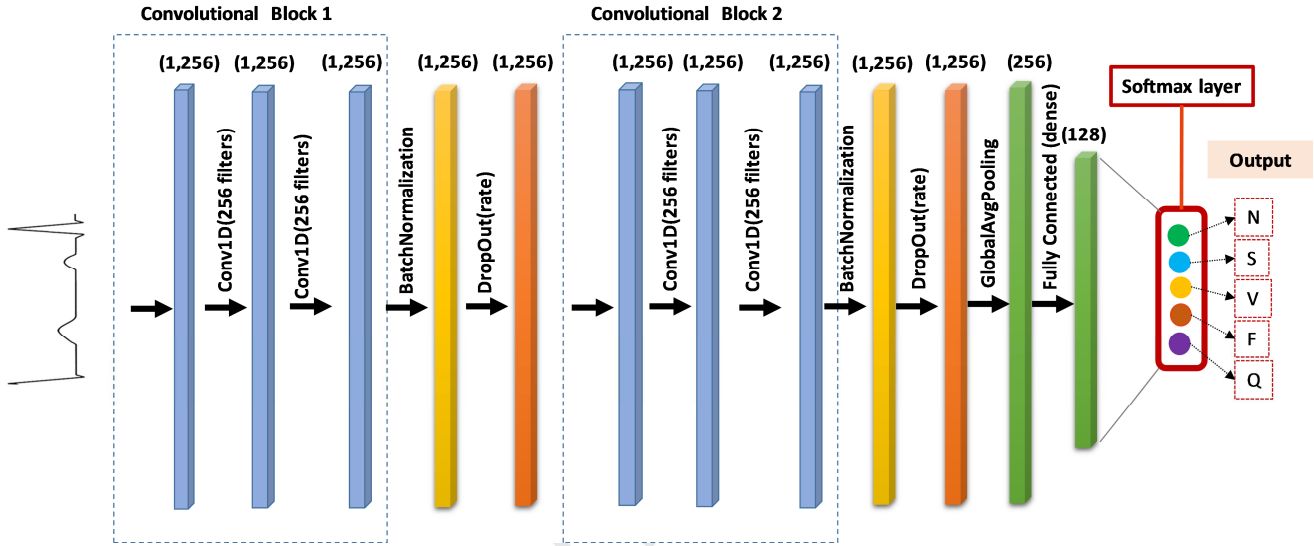


Fig. 3. Architecture of proposed convolutional neural network model.

At the end of the classification process, a dense layer preceded by a global average pooling method [46] is employed to revert back the feature map to a 1D tensor. This is because convolutional operations require two-dimensional tensors even if one of the dimension sizes is 1. The softmax activation function [47] is used to transform the outputs of the classifier into probabilities for each arrhythmia class. In the optimization phase, we used the Adam optimization algorithm and applied the focal loss instead of the regular loss function to reduce the negative impact of imbalanced data.

4. Experimental Results and Discussion

To evaluate the effectiveness of the focal loss approach, we compared the performance of the proposed deep CNN model with and without focal loss. We also conducted experiments to evaluate the impact of the batch normalization and dropout layers. The results were compared with those of recent studies to examine the performance of the proposed method.

4.1. Dataset description

In this study, we used the MIT-BIH arrhythmia database [48] and the St. Petersburg 12-lead arrhythmia database (INCART) [49]. MIT-BIH is a popular database used in similar studies [50] and is freely accessible from the Physionet website. This database is composed of two ECG recording channels sampled at 360 Hz that were obtained from 47 volunteer subjects (men and women) aged between 23 and 89 years. The records are identified by unique numbers, as illustrated in Table 2. They are divided into two parts: DS1 for training and DS2 for testing, with 22 records in each set. Each heartbeat was labeled by two cardiologists into 19 classes.

Table 2. Training and testing portions of MIT-BIH dataset.

Dataset name	Records included											Number of records
DS1 (training set)	101	106	108	109	112	114	115	116	118	119	122	22
	124	201	203	205	207	208	209	215	220	223	230	



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DS2 (test set)	100	103	105	111	113	117	121	123	200	202	210	
	212	213	214	219	221	222	228	231	232	233	234	22

INCART is another popular dataset also freely accessible from the Physionet website. This database consists of 75 annotated recordings extracted from 32 Holter records. Each record is 30 minutes long and contains 12 standard leads. Each lead is sampled at 257 Hz, with gains varying from 250 to 1,100 analog-to-digital converter units per millivolt. The reference annotation files contain over 175,000 beat annotations. Each heartbeat is categorized into 19 classes. The class annotations were produced by an automatic algorithm and then manually corrected.

Originally, 19 categories were used to identify the type of ECG heartbeat. Later, these categories were regrouped by the AAMI into five main classes [10]. Table 3 presents the five ECG heartbeat classes according to the AAMI standard: (N) normal beat, (S) atrial premature and supraventricular ectopic beat, (V) premature ventricular contraction and ventricular ectopic beat, (F) fusion of ventricular and normal beats, and (Q) unclassifiable or unknown beat.

Table 3. Mapping of MIT-BIH arrhythmia types to Association for Advancement of Medical Instrumentation (AAMI) heartbeat classes.

AAMI categories	Description	MIT-BIH arrhythmias (heartbeat types)
N	Normal beat	Normal beat (NOR) Atrial escape beat (AE) Nodal (junctional) escape beat (NE) Left bundle branch block beat (LBBB) Right bundle branch block beat (RBBB)
S	Supraventricular ectopic beat	Atrial premature beat (AP) Nodal (junctional) premature beat (NP) Supraventricular premature beat (SP) Aberrated atrial premature beat (AAP)
V	Ventricular ectopic beat	Ventricular escape beat (VE) Premature ventricular contraction (PVC)
F	Fusion beat	Fusion of ventricular and normal beat (FVN)
Q	Unknown beat	Paced beat (P) Unclassifiable beat (U) Fusion of paced and normal beat (FPN)

Table 4 presents the number of heartbeats in each class (N, S, V, F, and Q) for both datasets. This table indicates that both datasets were highly imbalanced. In addition, the MIT-BIH dataset was already divided by researchers into two sets: DS1, used as a training set, and DS2, used as a test set. A total of 22 records were included in the DS1 and DS2 sets, where the exact list of records included in each set is provided in Table 2. For the INCART dataset, there was no predefined division into training and test sets; thus, in this study, we randomly selected 50% of the data for training and the remaining 50% for testing.

Table 4. Distribution of heartbeats among five classes (N, S, V, F, and Q) for MIT-BIH and INCART datasets.

MIT-BIH Dataset				INCART dataset			
Class	Training (DS1)	Test (DS2)	Total	Class	Training (50%)	Test (50%)	Total
N	72,471	18,118	90,589	N	76,901	76,717	153,618
S	2,223	556	2,779	S	974	985	1,959
V	5,788	1,448	7,236	V	9,938	10,103	20,041
F	641	162	803	F	107	112	219
Q	6,431	1,608	8,039	Q	4	3	7
Number of heartbeats	87,554	21,892	109,446	Number of heartbeats	87,924	87,920	175,844



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Number of records	22	22	44
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Section 3.1 presents our proposal to segment ECG signals into heartbeats belonging to one of five classes in accordance with the MIT-BIH or INCART dataset. Figure 4 illustrates the results of our segmentation method by plotting the resultant heartbeats from the MIT-BIH dataset. However, due to the enormous number of heartbeats, we only used the first 50 heartbeats from the training dataset (DS1). Figure 4 presents five plots corresponding to the N, S, V, F, and Q classes. The average heartbeat is also plotted in black. A heartbeat starts at an R-peak and continues for 1.2 times the median RR time interval in a 10-s window. Figure 4 demonstrates that the location of the QRS complex varies and appears to be correlated with the class type. This method is simple and effective, as it ensures that most of the time, the heartbeat contains features that are important for correct classification, such as the RR intervals, PR interval, and QT interval [12-15].

In addition, all heartbeats are normalized by subtracting the average amplitude of the heartbeat and dividing by the standard deviation (also in relation amplitude of the heartbeat). The plots in Fig. 4 reveal slightly less intra-class variability for both the F and Q classes. However, these classes still have overlap with other classes. In fact, this inter-class overlap, or interference, is also present between all classes, which represents the main cause of ECG classification difficulty. The same segmentation process was also performed for the INCART dataset and similar results are achieved.

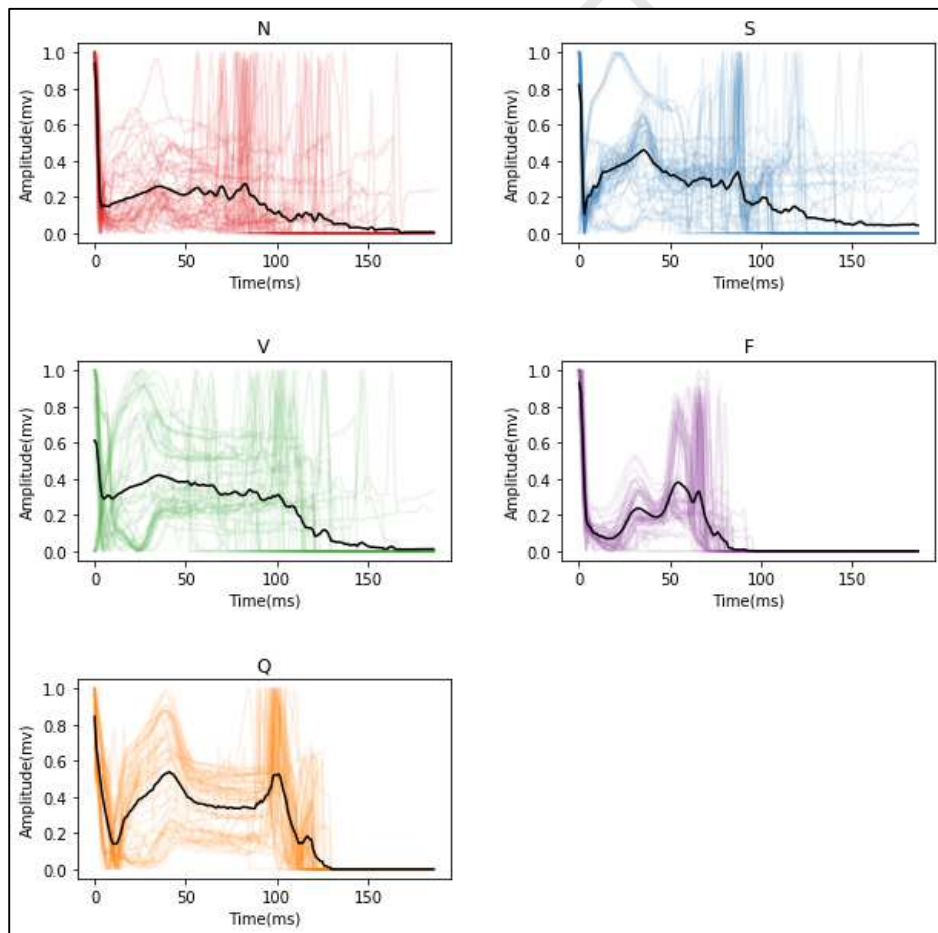


Fig. 4. Plot of the first 50 heartbeats from N, S, V, F, and Q classes as well as their average (black curve).



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4.2. Performance metrics

To evaluate the performance of our CNN model, we used five basic metrics: accuracy, area under the curve (AUC), F1-score, precision, and recall. The F1-score, AUC, precision, and recall metrics can be measured per class as well as for all scores. The performance metrics are defined based on the content of the confusion matrix (see Table 5). For a class k , let us define the following quantities as parameters used in score formulas (Eq. (12)):

$$\begin{aligned}
 TP_k &= T_{kk} \\
 TN_k &= \left(\sum_k T_{kk} \right) - T_{kk} \\
 FP_k &= \left(\sum_j T_{kj} \right) - T_{kk} \\
 FN_k &= \left(\sum_i T_{ik} \right) - T_{kk}
 \end{aligned} \tag{12}$$

Table 5. Definition of confusion matrix and parameters used in score formulas.

		Predicted classification					
		N(0)	S(1)	V(2)	F(3)	Q(4)	Total
True classifications	N	T_{00}	T_{10}	T_{20}	T_{30}	T_{40}	$\sum T_{i0}$
	S	T_{01}	T_{11}	T_{21}	T_{31}	T_{41}	$\sum T_{i1}$
	V	T_{02}	T_{12}	T_{22}	T_{32}	T_{42}	$\sum T_{i2}$
	F	T_{03}	T_{13}	T_{23}	T_{33}	T_{43}	$\sum T_{i3}$
	Q	T_{04}	T_{14}	T_{24}	T_{34}	T_{44}	$\sum T_{i4}$
	Total	$\sum T_{0j}$	$\sum T_{1j}$	$\sum T_{2j}$	$\sum T_{3j}$	$\sum T_{4j}$	$\sum \sum T_{ij}$

Now, the performance metrics are defined in two terms: (a) for individual classes and (b) as an overall metric:

- **Precision**, also called positive predictivity, is defined as the quotient of the true positive observations and the total predicted positive samples. A high precision percentage corresponds to a low false positive rate. Equation (13) computes the precision metric:

$$P_k = \frac{TP_k}{TP_k + FP_k} \tag{13}$$

The overall precision is computed as follows:

$$\text{Precision (+P)} = \frac{\sum TP_k}{(\sum TP_k) + (\sum FP_k)} \tag{14}$$

- **Recall**, also called sensitivity, measures the fraction of correctly predicted positive samples among the true positive and false negative observations within the class. The recall determines how many true-positive-labeled beats were found by the model, as expressed by Eq. (15):



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$$R_k = \frac{TP_k}{TP_k + FN_k} \quad (15)$$

The overall recall is computed as:

$$\text{Recall (Sensitivity)} = \frac{\sum TP_k}{(\sum TP_k) + (\sum FN_k)} \quad (16)$$

- **Specificity** measures the model's ability to predict true negatives of each available category. It is the fraction of correctly predicted negative samples among the true negative and false positive observations within the class. The specificity determines how many true-negative-labeled beats were found by the model, as illustrated by Eq. (17):

$$SPE_k = \frac{TN_k}{TN_k + FP_k} \quad (17)$$

The overall specificity is computed as follows:

$$\text{Specificity} = \frac{\sum TN_k}{(\sum TN_k) + (\sum FP_k)} \quad (18)$$

- **F1-score** is a crucial metric estimating the balance between the recall and precision metrics. In case of imbalanced classes, the F1-score is the most suitable metric to evaluate the performance of the classification model. Given a confusion matrix, as presented in Table 4, the F1-score is computed for each class as follows:

$$F1_k = 2 \times \frac{(R_k \times P_k)}{(R_k + P_k)} \quad (19)$$

The overall F1-score can be computed as a simple or weighted average of F1-scores per class.

- **Area under the curve (AUC)** is the area under the receiver operating characteristic (ROC) curve [51], which is a two-dimensional plot of the true positive rate. It is also referred to as the recall or sensitivity (y-axis) versus the false positive rate (x-axis), as expressed by Eq. (20). The higher the AUC measure, the better the model's performance. A perfect model has an AUC close to 1.

$$FPR_k = \frac{FP_k}{FP_k + TN_k} \quad (20)$$

The AUC corresponds to the arithmetic mean of the sensitivity (recall) and specificity values. In other words, we can calculate AUC_k for each class using the confusion matrix according to Eq. (21) [52]:

$$AUC_k = \frac{1}{2} (FPR_k + SPE_k) = \frac{1}{2} \left(\frac{FP_k}{FP_k + TN_k} + \frac{TN_k}{TN_k + FP_k} \right) \quad (21)$$

The overall AUC measure can be computed as a simple or weighted average of AUC_k per class.

- **Accuracy** is the most common performance measure expressed as the ratio of correct predictions to the total number of input observations. The overall accuracy is given by Eq. (22) as follows:

$$ACC = \frac{\sum T_{kk}}{\sum \sum T_{ij}} \quad (22)$$

Accuracy is the most commonly used metric to evaluate the performance and effectiveness of ECG classification methods. However, because heartbeat categories are very imbalanced, the accuracy may be significantly distorted by the main category (class N). Thus, the precision, recall, and F1-score should also be used as metrics to evaluate the performance and effectiveness of the model. An effective ECG heartbeat classifier should achieve high values in precision, sensitivity, and F1-score all at the same time. It should be noted that these performance metrics are recommended by the AAMI standard to establish a uniform environment for results comparison.

4.3. Results



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The proposed classifier was trained using a Dell PC with Intel Core i7-3520M 2.90 GHz processors and 8 GHz of RAM. Our CNN model was developed using the Python (3.7) programming language [53] and the Keras deep learning library [54]. The training of the deep CNN model was performed using a batch size of 400 and a learning rate of 0.001 for the Adam optimizer. Furthermore, we fixed the probability value of the dropout to 0.5 and the parameters of the focal loss function to $\alpha = 0.25$ and $\gamma = 2$. These values have been demonstrated in the literature to be optimum values leading to the best performance [41]. We also performed sensitivity analysis for these parameters in our experimental study.

4.3.1. Results for MIT-BIH dataset

In the first experiment, we trained a model that had one layer in a convolutional block and observed the rate of convergence of the model with and without focal loss. The loss and accuracy values for each epoch are plotted in Fig. 5, leading to two observations. First, model 2 (with focal loss) started with a much lower loss value and converged much earlier than model 1 (without focal loss). Second, both models converged after 100 epochs of training; thus, the number of epochs for the remainder of the experiments was set to 100. Furthermore, the learning rate was reduced to 0.0001 after the 80th epoch to fine-tune the Adam optimizer for the optimal model during training. In addition, we began tracking and saving the model providing the lowest loss value. This reduced the variability in the final optimization result because it consistently reached almost the same local minimum of the loss function.

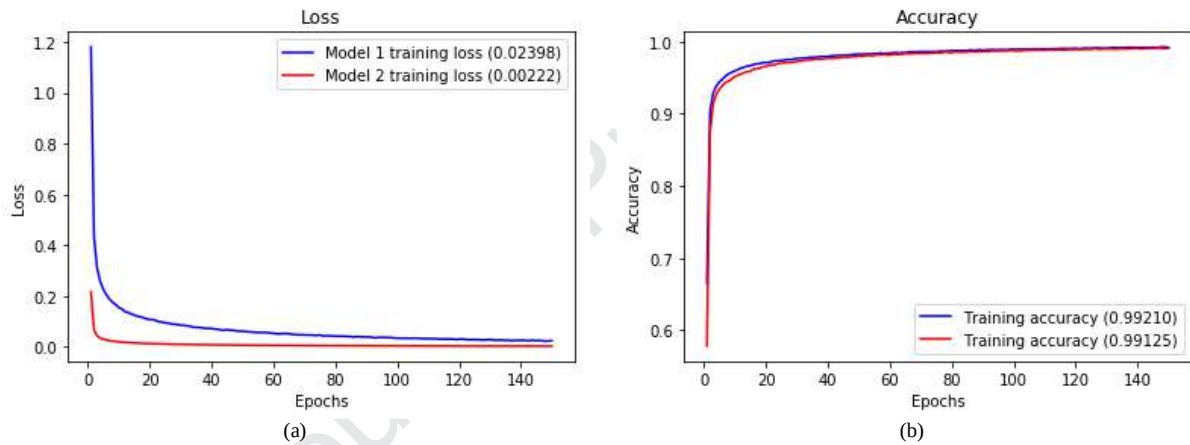


Fig. 5. Plots for accuracy and loss curves for model 1 (without focal loss) and model 2 (with focal loss), both using one convolutional layer in each block. (a) Loss curves, (b) accuracy curves.

In the second experiment, we were interested in evaluating the impact of the batch normalization layers on the proposed CNN models. The obtained accuracy results presented in Table 6 reveal that the batch normalization layers had a positive effect on the accuracy of both models. For model 2, the focal loss parameters were set to $\alpha = 0.25$ and $\gamma = 2$.

Table 6. Effect of batch normalization layers on accuracy.

	Without batch normalization	With batch normalization
Model 1 without focal loss	97.98	98.20
Model 2 with focal loss	98.30	98.41

Next, we conducted additional experiments to investigate the sensitivity of the accuracy to parameter α of the focal loss and dropout rate. Table 7 presents the overall accuracy of the proposed CNN model 2 (with focal loss) with variable values for the dropout rate and α parameter.

Table 7. Accuracy of proposed model with focal loss versus different values of dropout rate and α parameter.

	Values									
Dropout rate	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	



Accuracy	97.92	98.07	97.93	98.12	98.24	98.15	98.15	97.91	94.55		
α	0.10	0.20	0.25	0.30	0.35	0.40	0.50	0.60	0.70	0.80	0.90
Accuracy	98.00	98.33	98.44	98.25	98.11	97.54	98.07	98.02	98.05	98.09	97.93

The results in Table 7 are plotted in Fig. 6 for illustrative purposes. The dropout range [0.1, 0.7] provided almost the same accuracy result with a slight advantage for a dropout rate of 0.5. For the α parameter, 0.25 provided the best accuracy and was used as the optimal value for the remaining experiments.

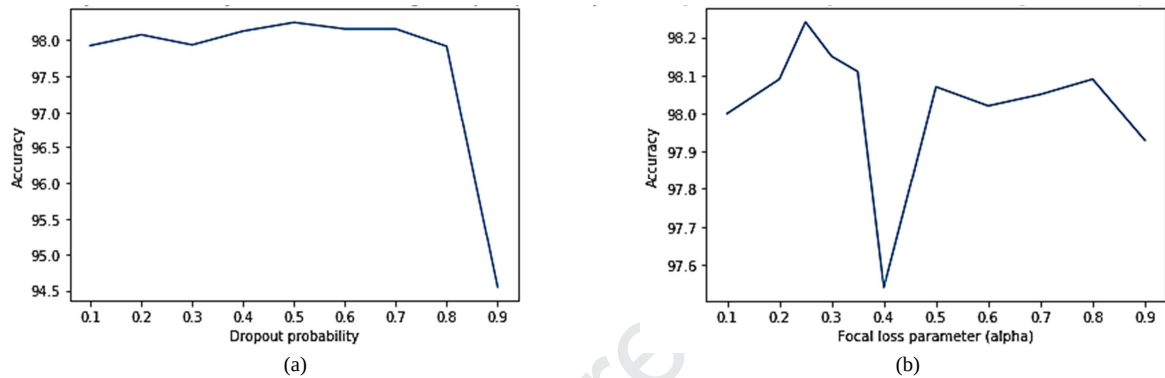


Fig. 6. Effect of dropout probability and α parameter. (a) Accuracy versus dropout probability; (b) accuracy versus α value.

We then investigated the effect of using focal loss together with the proposed CNN model. As previously fixed, the focal loss parameters were $\alpha = 0.25$ and $\gamma = 2$. Figure 7 presents the confusion matrix [55] for an example trial of both proposed CNN models 1 and 2 (without and with focal loss). The results indicate that the correct classification of the minority classes increased from 425 to 433 for the S class and from 122 to 133 for the F class. Even for the Q class, the number of correctly classified ECG heartbeats slightly increased from 1,581 to 1,584. This was due to the focal loss approach, which places more focus on the minority classes. However, the algorithm also misclassified some ECG heartbeats from the normal class as belonging to one of the minority classes (note that the number of correctly classified normal ECG heartbeats decreased from 18,051 to 18,025). For example, the number of normal ECG heartbeats that were incorrectly classified increased from 35 to 49 as class S, from 24 to 34 as class V, and from 4 to 6 as class F.

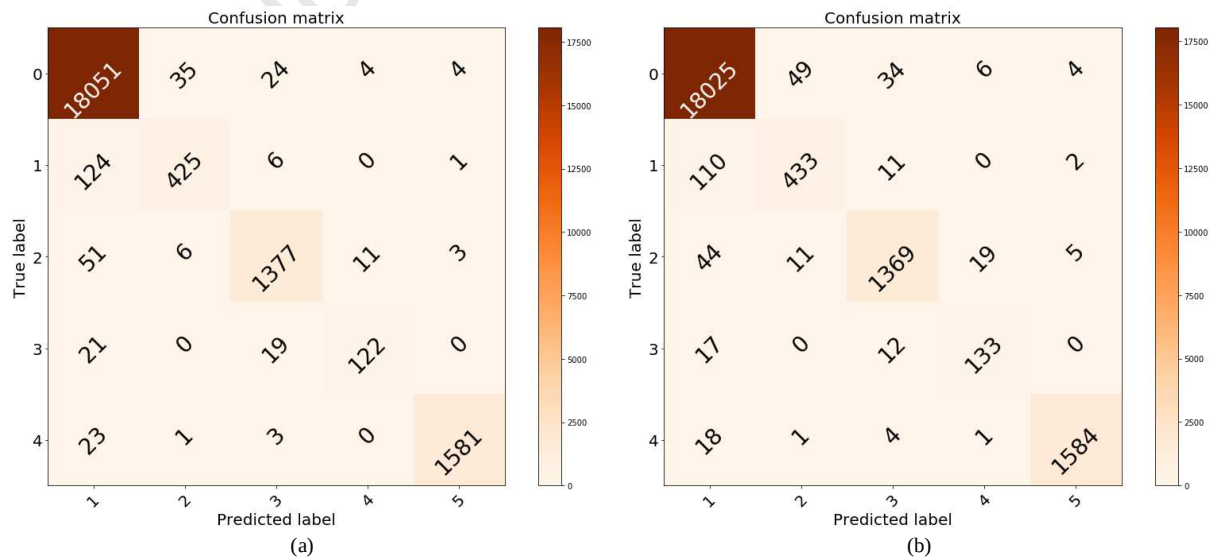


Fig. 7. Confusion matrix of proposed convolutional neural network model for an example trial. (a) Without focal loss, (b) with focal loss.



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However, Table 8 demonstrates that the results of the precision, recall, and F1-score for the normal class all increased. In other words, even though the number of correctly classified normal ECG heartbeats decreased (from 18,051 to 18,025), the number of abnormal heartbeats incorrectly classified as normal decreased even more. For minority classes S and F, Table 8 also indicates that the F1-score increased more significantly (from 80.89% to 82.48% for the S class and from 80.22% to 82.87% for the F class). Table 8 also presents the average F1-score, weighted average F1-score, weighted average AUC score, and accuracy. For all of these measures, the CNN model with focal loss performed better than the model without focal loss.

Table 8. Sample results of proposed model per class for MIT-BIH dataset without focal loss and with focal loss (focal loss parameters: $\alpha = 0.25$ and $\gamma = 2$).

Class	Without focal loss			With focal loss		
	Precision	Recall	F1 score	Precision	Recall	F1 score
N	98.81 $\pm$ 0.10	99.30 $\pm$ 0.27	99.05 $\pm$ 0.11	98.96 $\pm$ 0.03	99.49 $\pm$ 0.05	99.22 $\pm$ 0.03
S	85.39 $\pm$ 4.88	77.09 $\pm$ 1.73	80.89 $\pm$ 1.40	87.65 $\pm$ 1.51	77.88 $\pm$ 0.42	82.48 $\pm$ 0.72
V	95.81 $\pm$ 1.31	93.49 $\pm$ 0.66	94.63 $\pm$ 0.46	95.73 $\pm$ 0.30	94.54 $\pm$ 0.29	95.14 $\pm$ 0.12
F	81.11 $\pm$ 3.44	79.63 $\pm$ 3.68	80.22 $\pm$ 0.92	83.65 $\pm$ 0.89	82.10 $\pm$ 1.21	82.87 $\pm$ 0.79
Q	98.52 $\pm$ 0.76	98.33 $\pm$ 0.23	98.43 $\pm$ 0.36	99.31 $\pm$ 0.10	98.51 $\pm$ 0.15	98.91 $\pm$ 0.12
Average	91.93 $\pm$ 1.22	89.57 $\pm$ 0.60	90.64 $\pm$ 0.60	93.06 $\pm$ 0.30	90.50 $\pm$ 0.32	91.72 $\pm$ 0.24
Weighted average	98.11 $\pm$ 0.17	98.14 $\pm$ 0.19	98.11 $\pm$ 0.18	98.37 $\pm$ 0.06	98.41 $\pm$ 0.06	98.38 $\pm$ 0.05
Weighted AUC			99.02 $\pm$ 0.07			98.86 $\pm$ 0.10
Accuracy			98.14 $\pm$ 0.19			98.41 $\pm$ 0.06
Training time (s)			259.84			272.75

4.3.2. Results for INCART dataset

In this subsection, we evaluate the performance of our proposed model using the INCART dataset. Table 9 summarizes the results in terms of the precision, recall, and F1-score for the models with and without focal loss. The obtained results are similar to those for the MIT-BIH dataset. The accuracy improved from 96.88 $\pm$ 0.10 to 97.07 $\pm$ 0.04. The more significant improvements occurred for the minority classes S and F, where the improvements were from 56.17 $\pm$ 1.19 to 57.16 $\pm$ 0.65 and from 13.19 $\pm$ 0.45 to 15.05 $\pm$ 2.70, respectively. However, the ECG heartbeats belonging to the Q class could not be classified correctly. The number of samples from this class was only seven, and of the seven samples, only four samples were included in the training set, making the dataset extremely imbalanced. The number of samples in the Q class was too low for learning the model in an appropriate way, which led to a very low probability of correct classification. Therefore, other solutions are required to resolve this problem.

Table 9. Sample results of proposed model per class for INCART dataset without focal loss and with focal loss (focal loss parameters: $\alpha = 0.25$ and $\gamma = 2$).

Class	Without focal loss			With focal loss		
	Precision	Recall	F1 score	Precision	Recall	F1 score
N	97.96 $\pm$ 0.05	98.59 $\pm$ 0.10	98.27 $\pm$ 0.05	97.98 $\pm$ 0.04	98.78 $\pm$ 0.06	98.38 $\pm$ 0.02
S	59.34 $\pm$ 3.17	53.47 $\pm$ 1.77	56.17 $\pm$ 1.19	64.32 $\pm$ 1.37	51.45 $\pm$ 0.25	57.16 $\pm$ 0.65
V	91.96 $\pm$ 0.68	88.78 $\pm$ 0.60	90.34 $\pm$ 0.35	92.71 $\pm$ 0.42	89.16 $\pm$ 0.27	90.90 $\pm$ 0.17
F	21.82 $\pm$ 3.21	9.64 $\pm$ 1.04	13.19 $\pm$ 0.45	23.58 $\pm$ 4.95	11.07 $\pm$ 1.84	15.05 $\pm$ 2.70
Q	0.00	0.00	0.00	0.00	0.00	0.00
Average	54.22 $\pm$ 1.04	50.10 $\pm$ 0.21	51.60 $\pm$ 0.26	55.72 $\pm$ 0.75	50.09 $\pm$ 0.37	52.30 $\pm$ 0.44
Weighted average	96.77 $\pm$ 0.09	96.88 $\pm$ 0.10	96.81 $\pm$ 0.09	96.93 $\pm$ 0.04	97.07 $\pm$ 0.04	96.98 $\pm$ 0.04
Weighted AUC			97.32 $\pm$ 0.11			97.35 $\pm$ 0.07



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Accuracy	96.88 ± 0.10	97.07 ± 0.04
Time (s)	186.74	184.92

4.3.3. Comparison with state-of-the-art methods

Table 10 presents a large-scale comparison of our proposed CNN model with state-of-the-art methods. The F1-score is considered an important performance metric, especially for imbalanced data, and our model outperformed other state-of-the-art methods. In terms of the accuracy metric, our proposed method exceeded all other methods except for three methods proposed by Sanninoa et al. [27], Yildirim et al. [25], Li et al. [31], Pandey et al. [33], and Chen et al. [35]. However, Sanninoa et al. [26] addressed only two classes, which is a relatively simpler problem. The methods proposed by Li et al. [31], Pandey et al. [33], and Chen et al. [34] achieved better accuracy, but our proposed method outperformed them in terms of the F1-score, average precision, and/or average recall. Finally, Yildirim et al. [25] used 70%, 15%, and 15% of the MIT-BIH ECG signals to train, validate, and test their model, respectively, which achieved an average 99% accuracy, precision, recall, and F1-score. In contrast, we used DS1 for training and DS2 for testing. In other words, Yildirim et al. [25] used a different and larger training set than we did. Furthermore, their proposed system was characterized by a high complexity of the deep learning compression model and a long time required to obtain the coded ECG features.

Figure 8 provides an additional representation of our results for greater clarity. Figure 8(a) presents the ROC [51] curves of the proposed CNN model for different classes. For all heartbeat classes, ROC increased within the random guess and exceeded 94% for all classes except for class N, where it reached 99%. Figure 8(b) presents the real and predicted heartbeat categories in the test samples. The true labels (in blue) and predicted labels (in light red) are almost mingled. In particular, class N appears as a continuous dark red line reflecting the rate of correct prediction, which was very high.

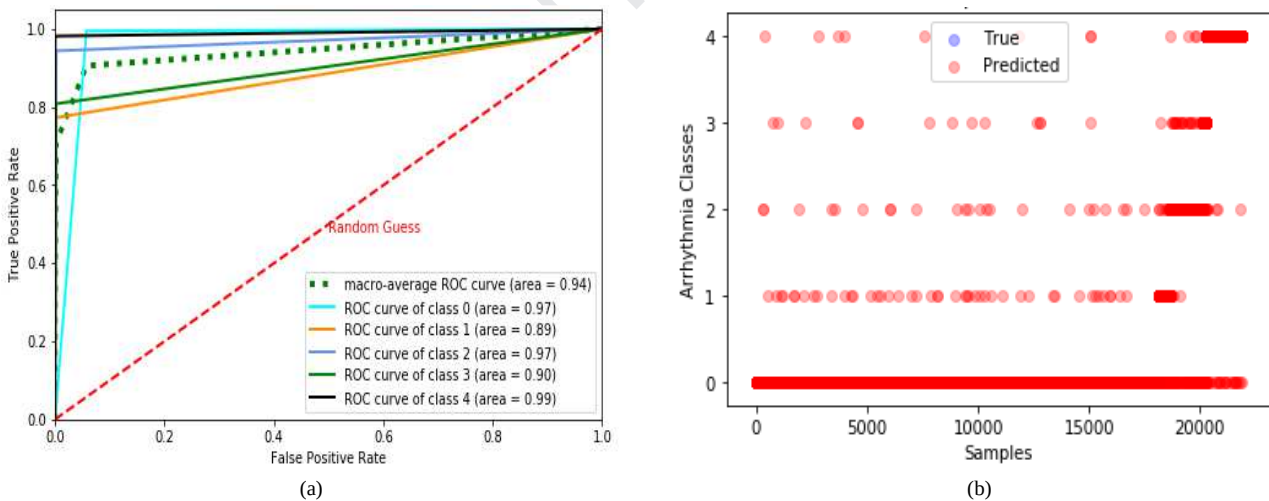


Fig. 8. Performance of proposed method in terms of (a) receiver operating characteristic (ROC) curve of convolutional neural network model, (b) plot of true versus predicted values.



Table 10. Comparison of performance of proposed convolutional neural network (CNN) model (with focal loss) with performance of state-of-the-art methods. (SEN: sensitivity)

Authors	Year	Approach	Num. of classes	Accuracy	AUC	F1-score	Precision	Recall (SEN)
Castilo et al. [4]	2012	Fusion of fuzzy-KNN and NN models	5	98.00%	-	-	-	-
Martis et al. [18]	2014	DWT and SVM	5	94.52%	-	-	-	98.61%
Li et al. [19]	2016	WPE and random forest	5	94.61%	-	-	-	-
Acharya et al. [21]	2017	Synthetic data and CNN	5	94.03%	-	97.28%	97.86%	96.71%
Garcia et al. [20]	2017	SVM and swarm optimization (PSO)	5	92.40%	-	-	-	81.10%
Amrani et al. [23]	2018	CNN and Q-Gaussian multi-class SVM	4	97.37%	-	-	96.43%	98.75%
Sanninoa et al. [27]	2018	Deep neural network	2	99.09%	-	-	-	98.55%
M. Kachuee et al. [29]	2018	Deep residual CNN	5	93.40%	-	-	-	-
Ramirez et al. [24]	2019	Hybrid neural network	10	93.80%	-	-	-	-
Yildirim et al. [25]	2019	CAE and LSTM	5	99.00%	-	-	-	99.00%
Marinho et al. [26]	2019	NB, SVM, MLP, and OPF	5	94.30%	-	-	-	-
S. Mousavi et al. [28]	2019	CNN and sequence-to-sequence model	5	96.18%	-	-	-	99.53%
Li et al. [31]	2020	Deep residual network	5	99.06%	-	-	96.76	93.21%
Pandey et al. [33]	2020	45 features + LSTM	5	99.37%	-	95.77%	96.73%	94.89%
Bidas et al. [35]	2020	Conditional entropy of ordinal patterns	5	93.62%	-	-	-	-
Chen et al. [34]	2020	CNN + bidirectional LSTM	5	99.56%	-	96.40%	-	-
Proposed method [ours]	2020	CNN + focal loss	5	98.41% $\pm$ 0.06	98.86 $\pm$ 0.10	98.38% $\pm$ 0.05	98.37% $\pm$ 0.06	98.41% $\pm$ 0.06



5. Conclusion

In this paper, a deep CNN model with focal loss is proposed to automatically classify ECG signals into five heartbeats categories (N, S, V, F, and Q) based on the AAMI standard. The focal loss approach reduces the impact of imbalanced heartbeat classes by focusing the loss on minority classes. We also propose a new method for segmenting ECG signals into separate heartbeats. This method has recently been proposed in the literature and, unlike previous standard methods, it defines a heartbeat as starting from an R-peak and continuing for a fixed number of samples. Thus, it does not require a resampling step, making it a simpler segmentation algorithm with fewer artificial modifications of the heartbeat.

To evaluate the proposed model, we used two ECG benchmarks, namely, the MIT-BIH and INCART datasets. The experimental results demonstrated the effectiveness of the focal loss technique in improving the classification accuracy of the method, especially for the minority classes (S and F). Our proposed CNN model outperformed state-of-the-art methods in ECG classification with a high overall accuracy as well as F1-score. Therefore, our ECG classifier is an effective tool for performing rapid and consistent arrhythmia diagnosis that can help cardiologists correctly identify heartbeat types.

In future work, the validity of the proposed model can be extended for other databases, such as the myocardial infarction dataset [56,57]. In addition, a cascaded approach can be investigated for training models in order to classify the ECG heartbeats into a normal class and all other classes. Another model can then be trained to classify abnormal ECG heartbeats into five separate classes. As a result, these two models can interact with more balanced classes.

DECLARATION OF INTEREST

The authors have no relevant conflicts of interest to disclose.

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AUTHOR CONTRIBUTION

Conceptualization, T. Romdhane, H. Alhichri, R. Ouni; Methodology, T. Romdhane, H. Alhichri; software, T. Romdhane; Formal analysis, T. Romdhane, H. Alhichri, R. Ouni, M. Atri; Investigation, T. Romdhane, H. Alhichri; Writing—original draft preparation, T. Romdhane; Writing—review and editing, T. Romdhane, H. Alhichri, R. Ouni, M. Atri; Supervision, M. Atri, R. Ouni; Project administration, R. Ouni; Funding acquisition, R. Ouni. All authors have approved the final article.

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Highlights (mandatory)

- Employing a new method for segmenting ECG signals into separate heartbeats that does not require any denoising, or resampling of ECG signals.
- Developing a deep CNN model for classifying heartbeats using the latest advanced designs and training techniques.
- Exploiting a novel type of loss function called *focal loss* which is effective in handling the imbalanced classes problem.
- The method is applied the MIT-BIH and INCART datasets to classify them in the five AAMI classes and achieved an average accuracy, AUC, F1-score, sensitivity and positive predictivity of 98.41%, 98.86, 98.38%, 98.37%, and 98.41% respectively.

DECLARATION OF INTEREST

The authors have no relevant conflicts of interest to disclose.

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